



- Tactical OPE Toolkit - Automating Location Intelligence for Special Operations Using AI



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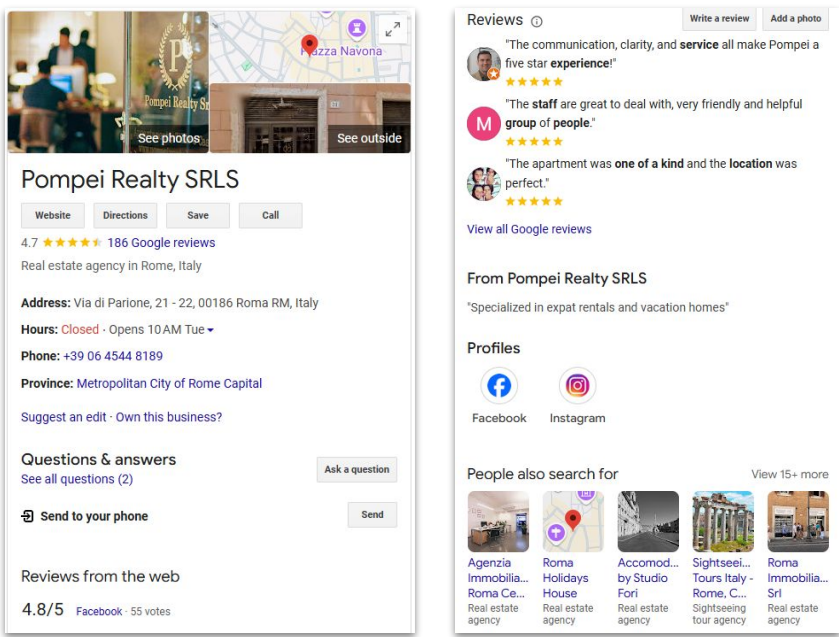
1. Background & Motivation

The Special Operations Forces (SOF) conducts worldwide special operations, often in denied or restricted territory. Understanding the operational environment is essential, and the **SOF analysts provide situational awareness through detailed analysis, which supports mission planning and execution**. While civil considerations can often be gathered through tools like Google Maps, **there is currently no automated method to efficiently extract and analyze this information**.

2. Problem Statement

How can we **retrieve location details** from Google Maps and generate an Excel, with **image analysis and review summarization** to enable operational planning and execution **while reducing the man hours and cost**?

3. Current Solution



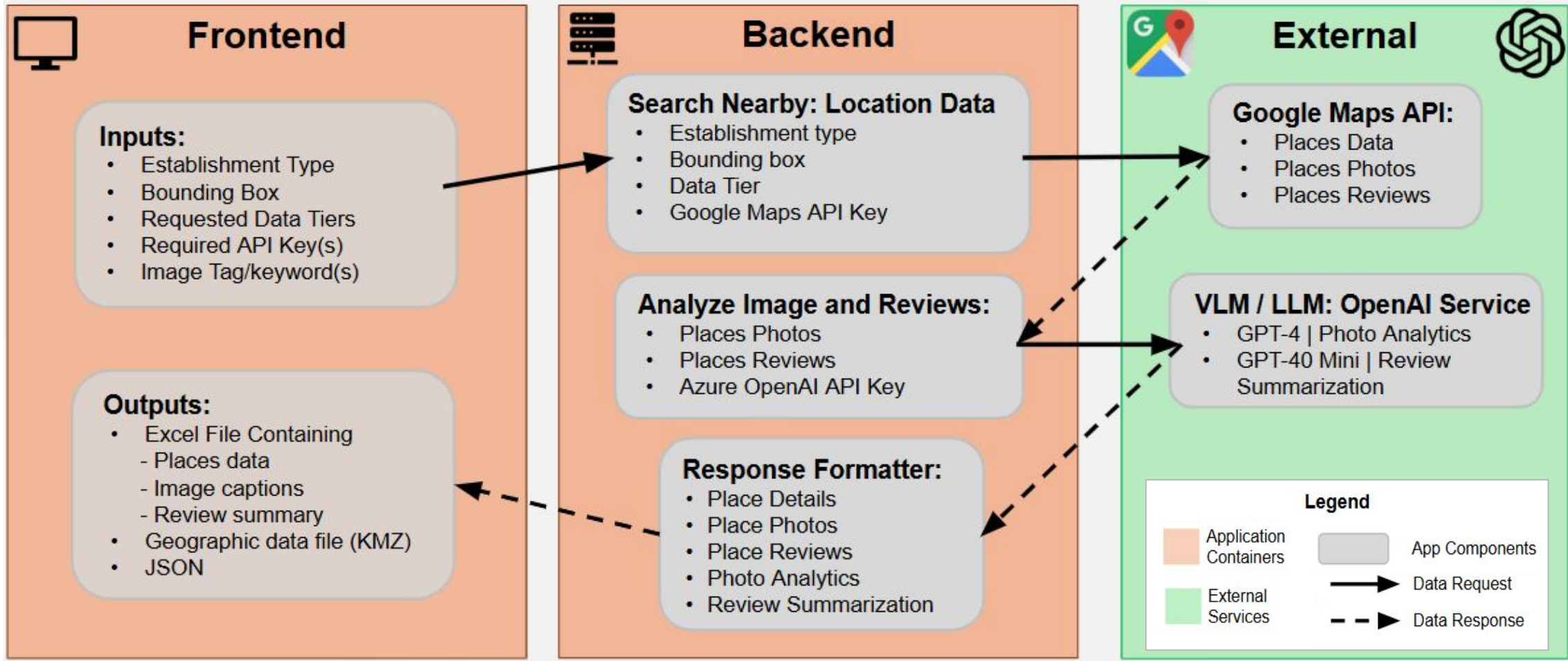
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photo	street_view	working_hours	location_link	email_1
https://lh5.googleusercontent.com/p/AF1QipMIWJupgzysLIGtKbKw3GLI2pa6Uti285bF3ak=w800-h500-k-no	https://lh5.googleusercontent.com/p/AF1QipMIWJupgzysLIGtKbKw3GLI2pa6Uti285bF3ak=w800-h500-k-no	["Monday": "10AM\u00136PM", "Tuesday": "10AM\u00136PM", "Wednesday": "10AM\u00136PM", "Thursday": "10AM\u00136PM", "Friday": "10AM\u00136PM", "Saturday": "10AM\u00131PM", "Sunday": "Closed"]	https://www.google.com/maps/place/Pompei+Realty+SRLS/@41.8984824,12.4707807,14z/data=!4m8!1m2!1m1!1sPompei+Realty+SRLS!1s1s0x132f600b128f44ed:0x19d0fc186211a3dc18m2!3d41.8984824!4d12.470780699999999	info@realnynproperties.com

Manual Process Overview

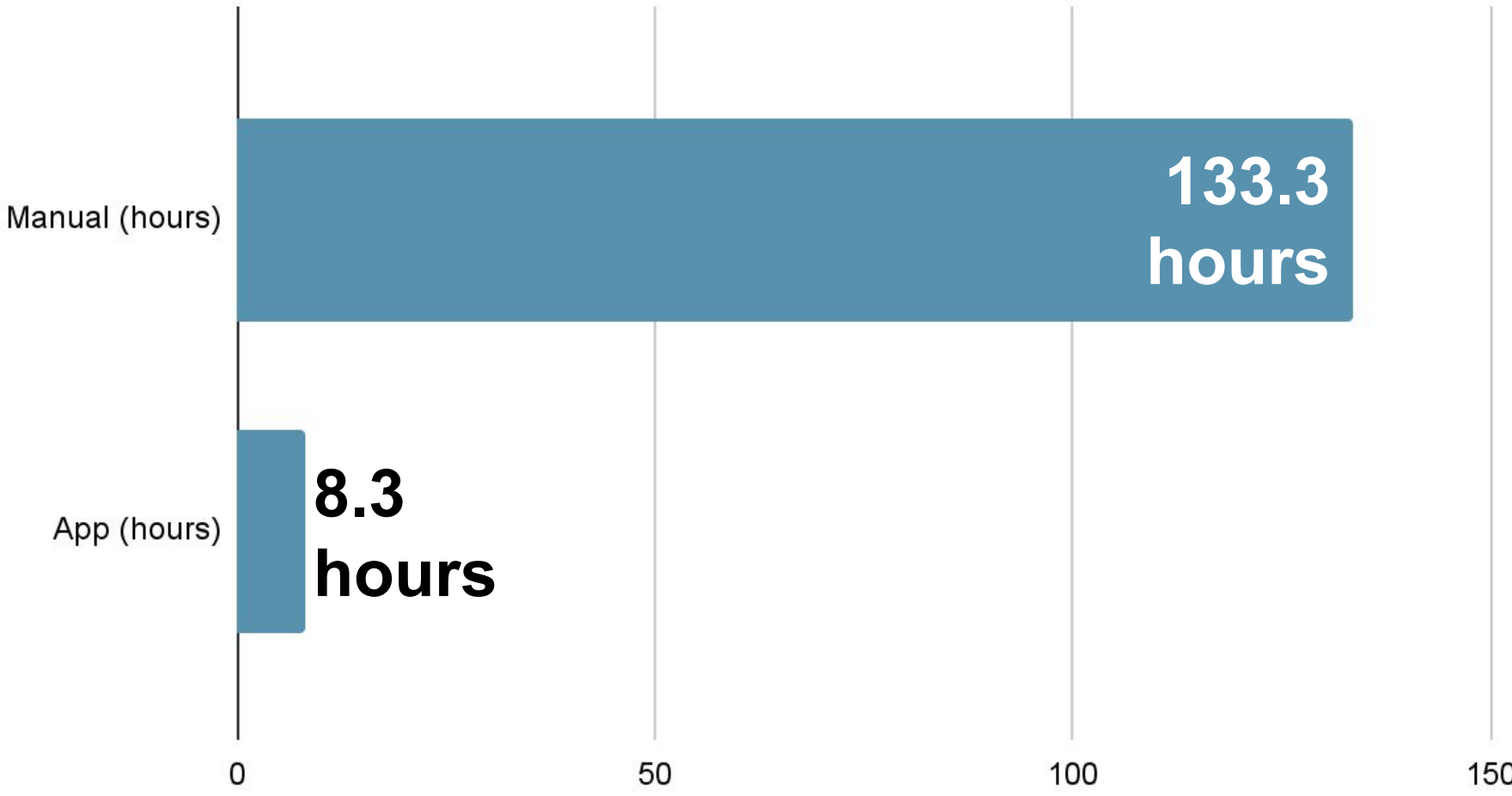
- Open Google Maps and select the location
- Copy and paste the data into the Excel file
- Triage the reviews and photos; author a summary
- Highlight key features in reviews and photos
- 20 min process time per locations
- 8 - 40 hours to complete each report
- Missions require a 24 hour turn around
- Estimated 400 locations per month = 133.3 hours

4. Solution Design



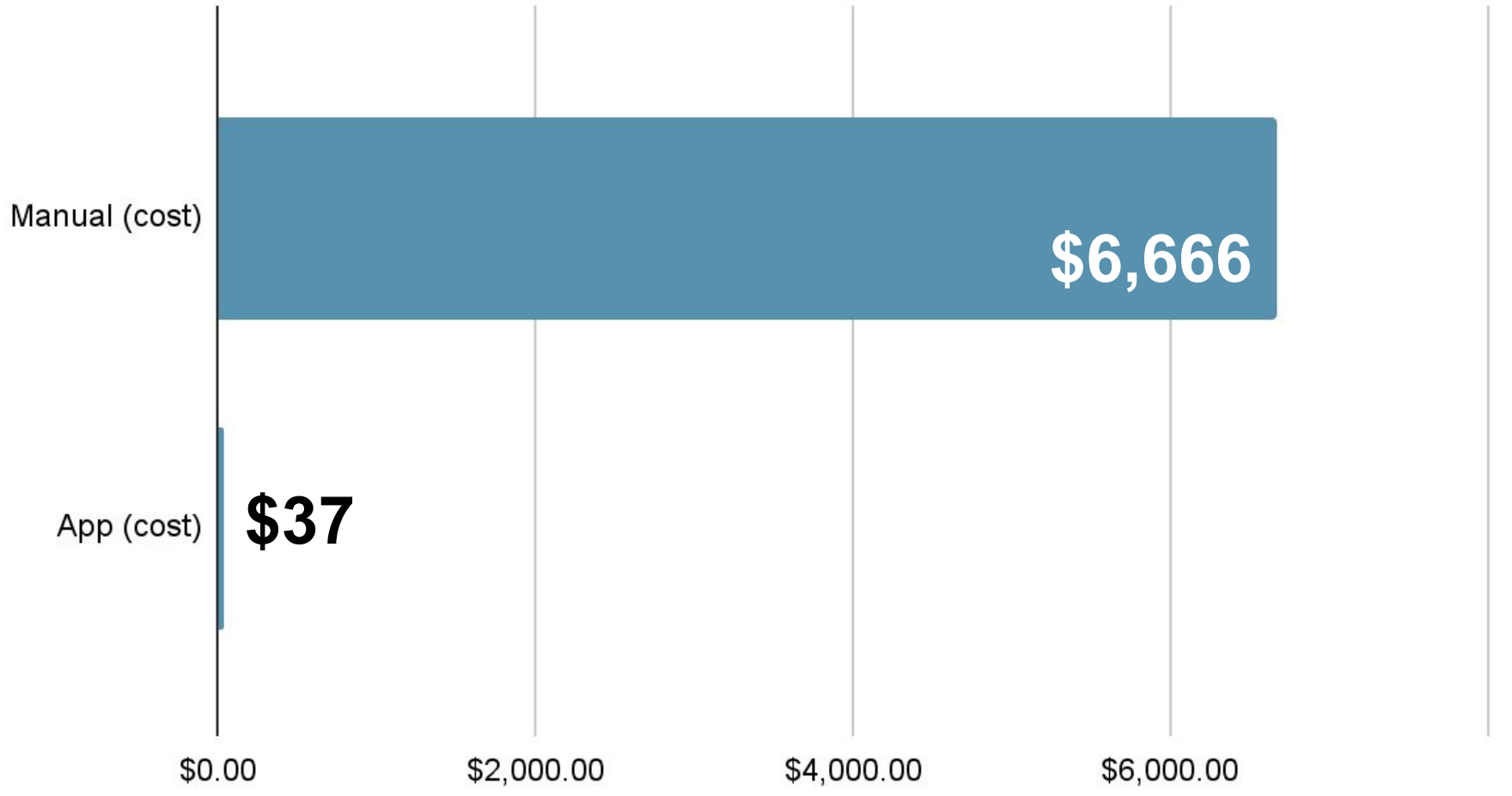
6. Impact

Hours per month (400 locations)



94% reduction in schedule;
saving 125 hours per month

Cost per month (400 locations)



99.4% reduction in cost;
saving \$6,629 per month

5. Application Implementation

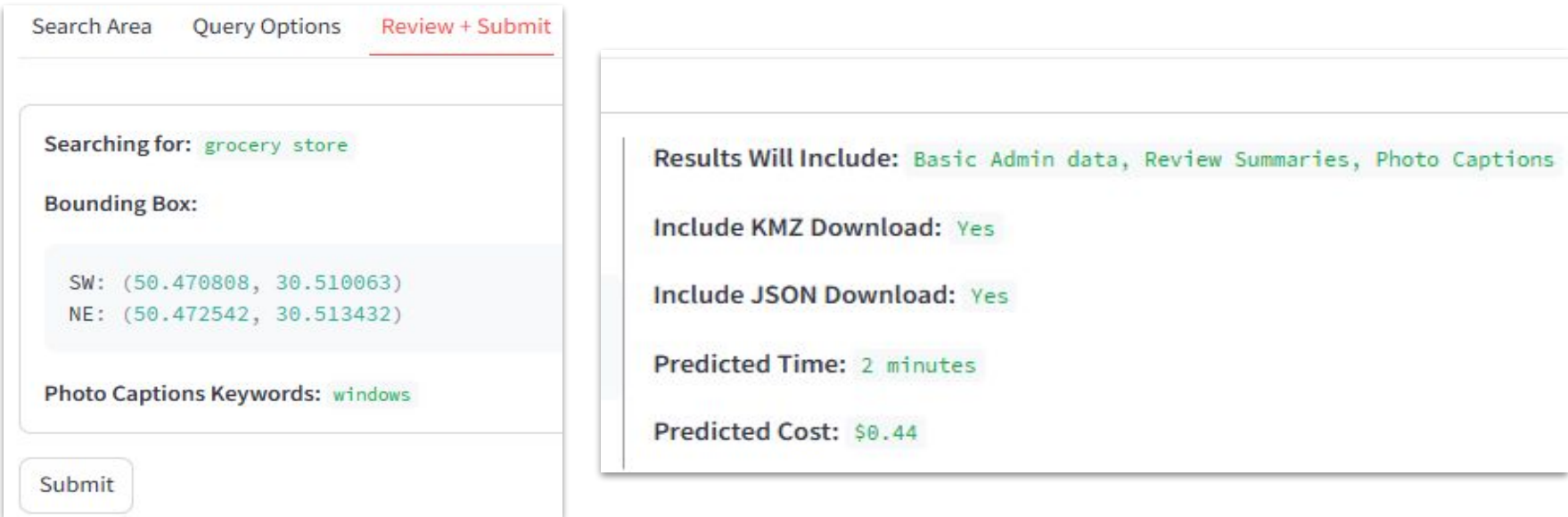
Searching

- Drop pins
- Draw bounding box
- Add bounding box

Querying

- Establishment search term
- Data tiers (basic, review, photos)
- Outputs (Excel, KMZ, and JSON)
- API key handling
- Photo captioning keywords
- Time and cost prediction

Validating



7. Key Takeaways

- Reduction of 94% in schedule and 99.4% in cost
- Reduced the risk of human error
- Automated Excel generation
- Additional outputs: KMZ and JSON
- More information: AI summarized reviews & photos
- Accessible via the cloud or local host

For more information on the Tactical OPE Toolkit including a demonstration video and documentation - use the QR code



Supervisor - Eric Youn and Zach Murphy
AI Technicians 2024-2025 - Capstone Project